

Reviewer Pack — Coxeter-Weyl Group Action on Boolean Circuit Tensor Rank

Entry 5fc15b60077d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-28 14:05:22 UTC

Coxeter-Weyl Group Action on Boolean Circuit Tensor Rank

Entry ID: 5fc15b60077d **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 14:05:22 UTC

1. Conjecture

Field A (mathematical branch): Coxeter and Weyl Algebras **Field B** (complexity object): Boolean Circuit Tensor Rank

Statement:

For every boolean function f represented as a circuit of size s , the minimum rank of the tensor representing f in the Coxeter-Weyl algebra is upper bounded by a polynomial in s .

Rationale (proposer's reasoning):

Coxeter and Weyl algebras provide a framework to study invariant properties of objects under symmetry operations. The tensor rank of a boolean circuit could be an invariant that captures structural complexity, potentially revealing insights into the relationship between circuit size and computational hardness.

Taxonomy category: GCT_DET_PERM (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 1b97f44b74ac3662

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For boolean functions represented as circuits of size $s \leq 40$, the tensor rank in the Coxeter-Weyl algebra is upper bounded by s^k for some k .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
---------	---------	------------	------------------	------------------

4. Novelty audit

Verdict: NOVEL against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - circuit tensor rank AND Coxeter-Weyl algebra - Boolean function representation AND Weyl algebra AND tensor rank - tensor rank of Boolean circuits IN Coxeter algebra

Top relevant hits considered: - [http://arxiv.org/abs/hep-th/0610278v4] A Special Class of Rank 10 and 11 Coxeter Groups - [http://arxiv.org/abs/math/0405176v4] Quantized symplectic oscillator algebras of rank one - [http://arxiv.org/abs/1306.5684v3] New Large-Rank Nichols Algebras Over Nonabelian Groups With Commutator Subgroup Z_2 - [http://arxiv.org/abs/0711.0795v4] Finite-Dimensional Representations of Hyper Loop Algebras Over Non-Algebraically Closed Fields - [http://arxiv.org/abs/1311.5217v2] Representation theory of Mackey Lie algebras and their dense subalgebras - [http://arxiv.org/abs/2407.04826v1] Multi-strategy Based Quantum Cost Reduction of Quantum Boolean Circuits - [http://arxiv.org/abs/0804.0947v2] The Dynkin diagram cohomology of finite Coxeter groups

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def tensor_rank(circuit):
        n = len(circuit)
        if n == 1:
            return 1
        rank = 1
        for i in range(1, n):
            rank *= (i + 1)
        return rank

    def is_polynomial_bound(rank, s, k):
        return rank <= s**k

    max_n = 40
    instances_tested = 0
    total_rank = 0
    conjecture_holds = True
    counterexample = ""

    for n in range(5, max_n + 1, 5): # Sweep through 5, 10, 15, 20, 30, 40
```

```

circuit = generate_boolean_function(n)
rank = tensor_rank(circuit)
total_rank += rank
instances_tested += 1

k = 1 # Assuming the conjecture holds for k=1 initially
if not is_polynomial_bound(rank, n, k):
    conjecture_holds = False
    counterexample = f"n={n}, rank={rank}"

mean_rank = total_rank / instances_tested
std_deviation = 0.0

return {
    "metric_name": "tensor_rank",
    "metric_value": mean_rank,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 97)) # Default to first 30 primes if

    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    std_deviation = math.sqrt(sum((r["metric_value"] - mean_rank) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std={std_deviation} support_fraction={support_f")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_rank} std={std_deviation} support_fraction={support_f")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample={result['counterexample']}" first_failing_seed)

```

6. Per-seed results

(no seed-level data — test crashed before producing *TRIAL:* lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e5190b5.py", line 71, in <module>
    result = run_trial(seed)
    ^^^^^^^^^^^^^^^^^

```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e5190b5.py", line 51, in run_trial
    counterexample = f"n={n}, rank={rank}"
                        ^^^^^^^
```

ValueError: Exceeds the limit (4300 digits) for integer string conversion; use sys.set_int_max_str_digits to increase the limit if you need to convert an integer of this size to a string.

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which prevents a definitive evaluation of the conjecture. | next: Re-run the test with increased limits for integer string conversion to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10464
2	preregistration	ollama_remote	glm4:latest	0	0	5406
3	novelty	ollama_remote	glm4:latest	0	0	4500
4	novelty	ollama_remote	glm4:latest	0	0	5720
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14593
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9386
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14692
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7996
9	judge	ollama_remote	glm4:latest	0	0	42743

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 115500 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/5fc15b60077d.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/5fc15b60077d.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/5fc15b60077d.tar.gz` (if generated)